mitrafield

# Explain it once.
# Every beat finds its engine.

An AI director that writes explainer videos and sends every beat to the engine that can show it truthfully: exact math, real maps, charts and diagrams where accuracy matters, Veo and Nano Banana where mood does.

Google Cloud AI Builder Cup 2026 · Theme: **Media, Content & Digital Experiences**

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THE PROBLEM

# Explainer videos are how the world learns.
# Making a correct one is still slow, expensive and manual.

## days

### of work per minute

A good explainer needs a script, research, animation, maps, charts, narration and editing. Teachers, NGOs and small creators cannot afford a studio.

## wrong

### numbers drawn by pixels

Text-to-video tools make beautiful clips, but they hallucinate. Coastlines are invented, figures are garbled, equations are squiggles. For learning, a wrong number is worse than none.

## one

### language, one format

Most explainers exist only in English and only as long-form. Re-making them as Shorts or in Hindi, Nepali or Bahasa means starting again.

THE SOLUTION

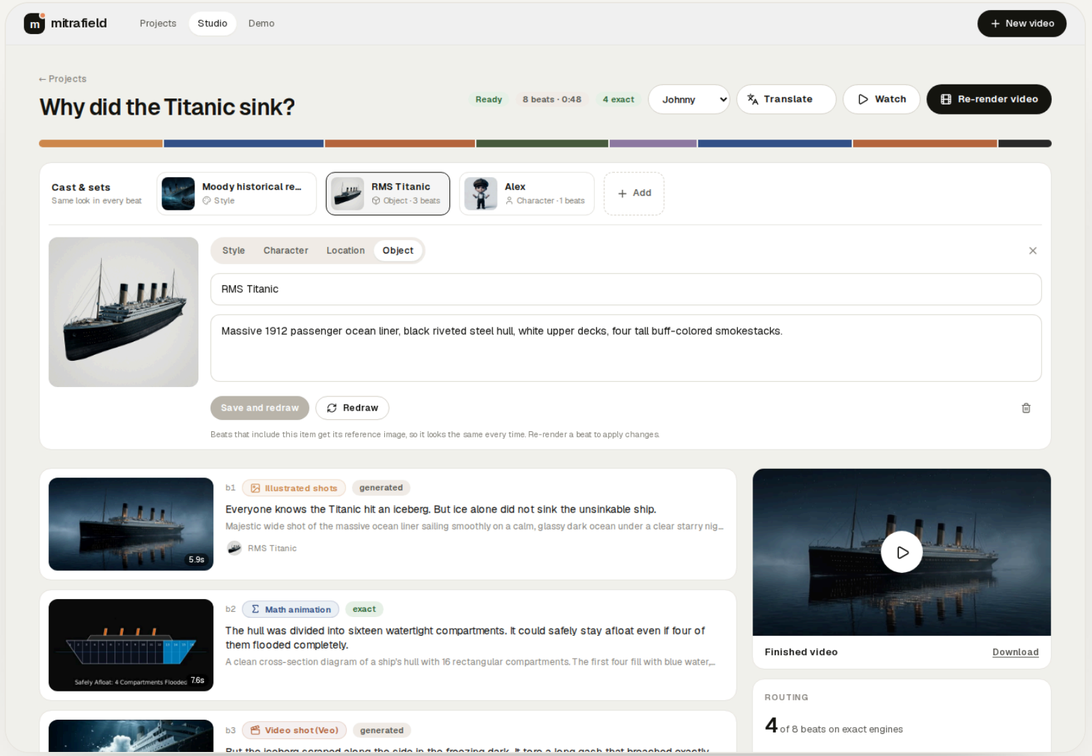
# A director agent with eleven engines and ten looks.
## Precise where it matters, generative where it helps.

You ask a question and add notes or PDFs. Gemini writes a storyboard for the ear, beat by beat, and routes each beat to the engine that can show it truthfully. Every beat is rendered to the exact length of its narration, checked, and assembled into a finished video with music and subtitles.

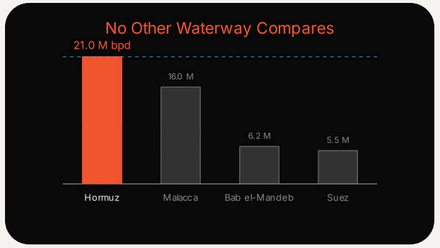
**exact** Motion scene · Math animation · Globe & maps · Data chart · Timeline · Flow diagram · Whiteboard · Kinetic type

**generated** Cinematic shots · Presenter · Veo video

Facts, numbers, maps and math are always drawn by code from real data. Ten looks (Cinematic, Editorial flat, Premium 3D, Ink & wash, Motion graphics, Archival, Woodblock, Paper collage, Isometric, Blueprint) restyle every engine at once.



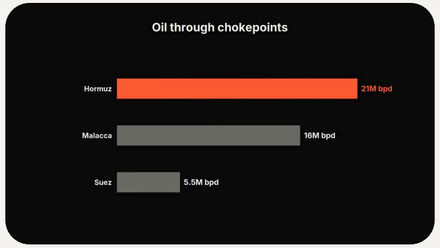
# Every engine plans a beat and renders it to the exact length.



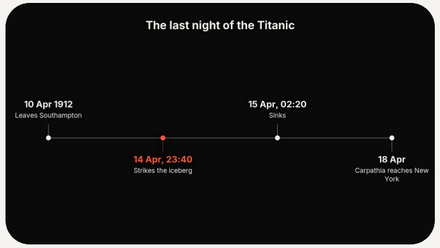

**Math animation**
Gemini writes Manim code, repairs errors, checks frames



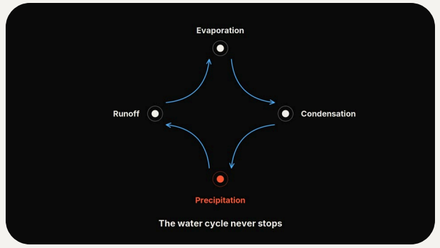

**Globe & maps**
Natural Earth borders, flights, pins, routes



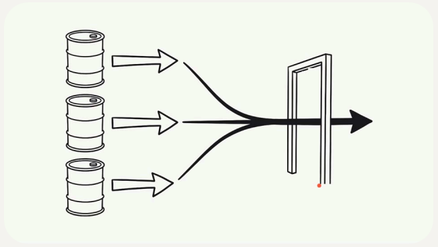

**Data chart**
Bars and trends from source numbers




**Timeline**
Events at their real dates




**Flow diagram**
Processes, cycles, trees



**Whiteboard**
Line art drawn on stroke by stroke




**Kinetic type**
Hooks and key numbers



**Cinematic shots**
Photographic stills in one grade



**Presenter**
One real-looking person, same face every beat

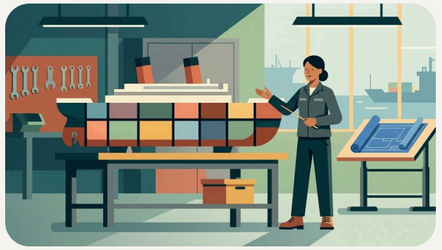


**Veo shots**
Hero motion, max two per video

# One story, ten looks. Professional in every one.



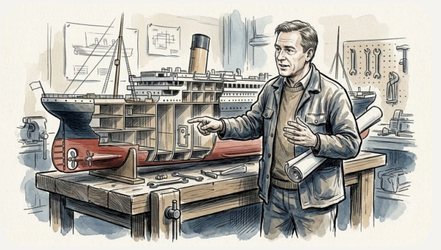
**Cinematic**



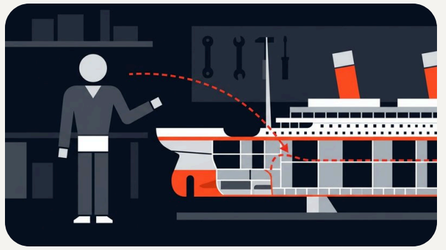
**Editorial flat**



**Premium 3D**



**Ink & wash**



**Motion graphics**



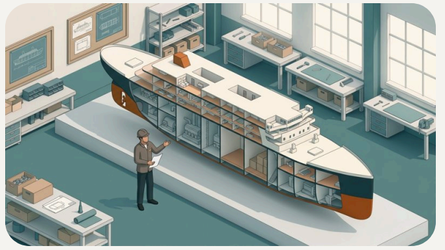
**Archival**



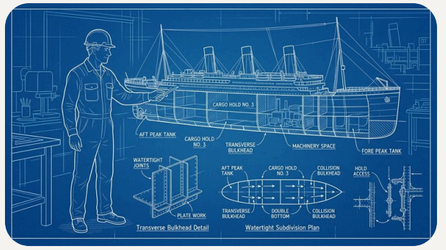
**Woodblock**



**Paper collage**



**Isometric**




**Blueprint**

A look sets the style of every generated image and its character rules, and gives code-drawn engines its palette and paper texture, so the whole video matches. Same subject shown in each.

HOW IT WORKS

# From a question to a finished video.
# You only make the calls.

| | | | |
|---|---|---|---|
| 01 | **Ask** | YOU | A question plus optional notes, PDFs or images. Format, length, language, narrator, audience. |
| 02 | **Storyboard + cast** | GEMINI 3.1 PRO | Narration beat by beat, an engine per beat, and the recurring cast: style, characters, objects, places. |
| 03 | **Narrate** | GEMINI TTS | Each line is voiced first, then transcribed and retaken if it drifts from the script. |
| 04 | **Render + self-check** | ENGINES | Beats render in parallel to their exact length. Manim code is repaired on error and inspected by Gemini vision. |
| 05 | **Edit one beat** | YOU + DIRECTOR | Rewrite a line, swap an engine, or say "show it on a map". Only that beat re-renders. |
| 06 | **Assemble + translate** | LYRIA · FFMPEG | Music ducked under the voice, -14 LUFS, subtitles. One-click versions in other languages reuse the visuals. |

# Define it once. The same ship in every shot.

The director defines the recurring ingredients with the storyboard. Each gets one reference image; every generated beat is drawn from the references it contains, and Veo animates an on-model first frame.



**RMS Titanic**
object · black hull, white decks, four buff funnels



**Thomas**
presenter · 1912 Harland and Wolff shipbuilder



**Moody historical realism**
style · applied to every generated beat



Beat 1 · illustrated still



Beat 3 · Veo video



Beat 7 · Veo video




Beat 5 · presenter

ARCHITECTURE ON GOOGLE CLOUD

# One Cloud Run service, Google models end to end.

**Workspace**
Next.js app: new video, studio, cast strip, inspector, projects, demo

**Your sources**
Notes, PDFs and images read by Gemini as the source of truth

→

**Cloud Run · FastAPI**
Director agent · per-beat pipeline (narrate → plan → render → fit) · content-hash cache · cost ledger · passcode for writes

**Code engines**
Manim + LaTeX, pycairo globe, charts, timeline, flow

**Assembly**
ffmpeg: fit, crossfade, mix, loudnorm, .srt

Cloud Storage mount for projects and media · Secret Manager for keys

→

**Gemini 3.1 Pro**
storyboard, routing, cast, rewrites, translation

**Gemini Flash**
engine planning, Manim code, vision QA, transcripts

**Nano Banana**
stills, characters, line art, cast references

**Veo 3.1**
hero shots, image-to-video from cast frames

**Gemini TTS · Lyria**
narration · topic-matched music bed

GEN AI IMPLEMENTATION

# Not one prompt. Five loops that check each other.

01

### Route

The director reads every engine's description and picks per beat, under rules: facts go to exact engines, Veo is capped at two.

02

### Code + repair

Gemini writes Manim for the exact duration. Render errors go back with the traceback until it compiles.

03

### Look + fix

Gemini vision inspects rendered frames for overlapping or cut-off text and asks for a corrected scene.

04

### Listen + retake

Every narration line is transcribed and retaken if words were added, dropped or garbled.

05

### Stay on model

Cast references feed every image and Veo first frame, so characters and subjects stay consistent.

Failures degrade gracefully: a refused Veo clip becomes a still, a failed engine becomes a title card, and the video still renders.

RESULTS FROM THE PROTOTYPE

# Finished explainers from a single prompt.

**10**

finished videos: geography, math, biology, history, tech, anatomy and earth science, in 3 looks and 3 languages

**2–9 min**

to render a 45–60 s video on one small server, unattended

**$0.72–1.70**

measured API cost per video, logged per call; a translation that reuses the shots cost $0.34

**56%**

of all beats rendered by exact engines, so facts are computed, not generated



Woodblock look · tsunami




Editorial flat · vaccine Short with a cast presenter




Hindi translation · labels re-rendered in Devanagari

IMPACT

# Correct explainers for everyone who teaches.
# In their language, in their format.

### Teachers and schools

Turn a lesson plan into a narrated video with accurate diagrams, then into a Short for revision. One teacher, minutes, a few dollars.

### NGOs and newsrooms

Explain a policy, a disease or a crisis with real maps and real numbers, cited on screen, and publish it in several JAPAC languages the same day.

### Creators

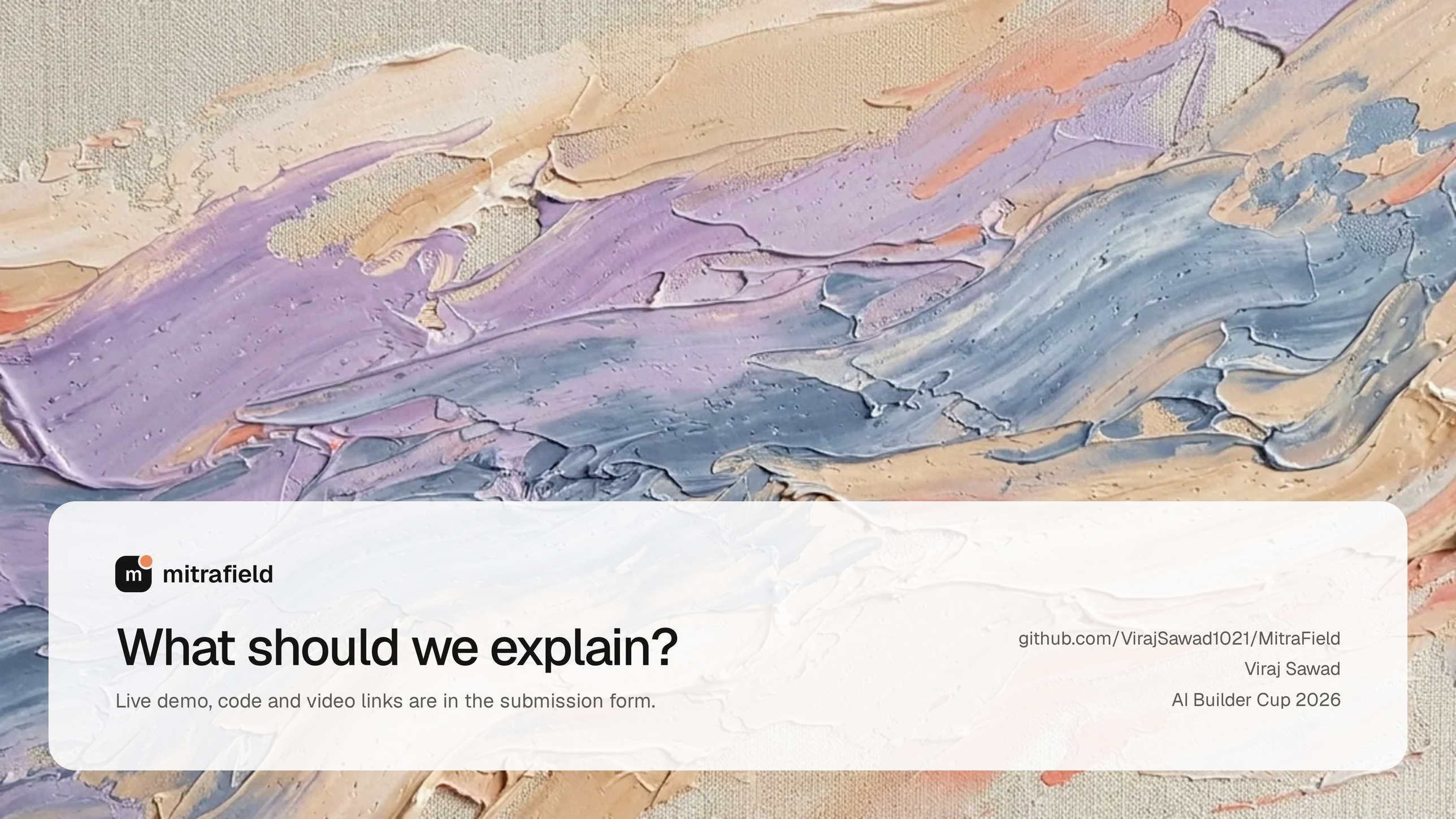
Run an explainer channel with a consistent presenter and style across every episode, long-form and Shorts from one workspace.

Misinformation angle: every number and map in a MitraField video is computed from data the user can see, not invented by a pixel model.

FEASIBILITY AND SCALE

# Built to run cheaply, and to grow.

| AREA | TODAY | NEXT |
| --- | --- | --- |
| **Compute** | One Cloud Run service, renders as in-process jobs, content-hash cache | Renders as Cloud Run jobs, one per beat, scale to zero |
| **Cost control** | Per-project ledger, Veo length fitted to the beat, draft mode on Veo Lite | Batch API for final renders, per-user budgets |
| **Styles** | Ten engines behind one contract, discovered automatically | Scale comparisons, source-quote cards, code walkthroughs, real archival footage |
| **Learning** | Audience and level steer the script | Learn mode: quizzes after the video, beats re-explained where viewers struggle, analytics in BigQuery |
| **Languages** | One-click translation that reuses the visuals | Cloud Translation glossaries, regional voices |



mitrafield

# What should we explain?

Live demo, code and video links are in the submission form.

github.com/VirajSawad1021/MitraField

Viraj Sawad

AI Builder Cup 2026